CS 599 – Project Proposal

Citation-Grounded LLM System for Academic Summarization and Question Answering

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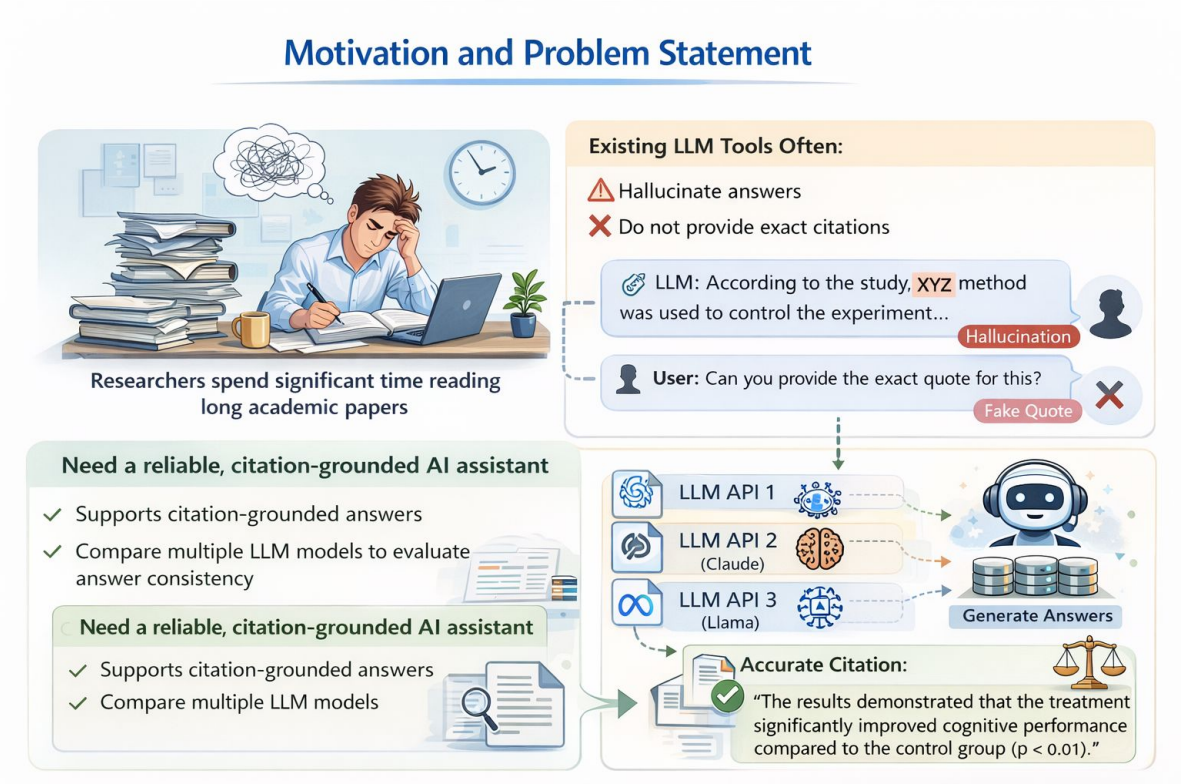
Motivation and Problem Statement

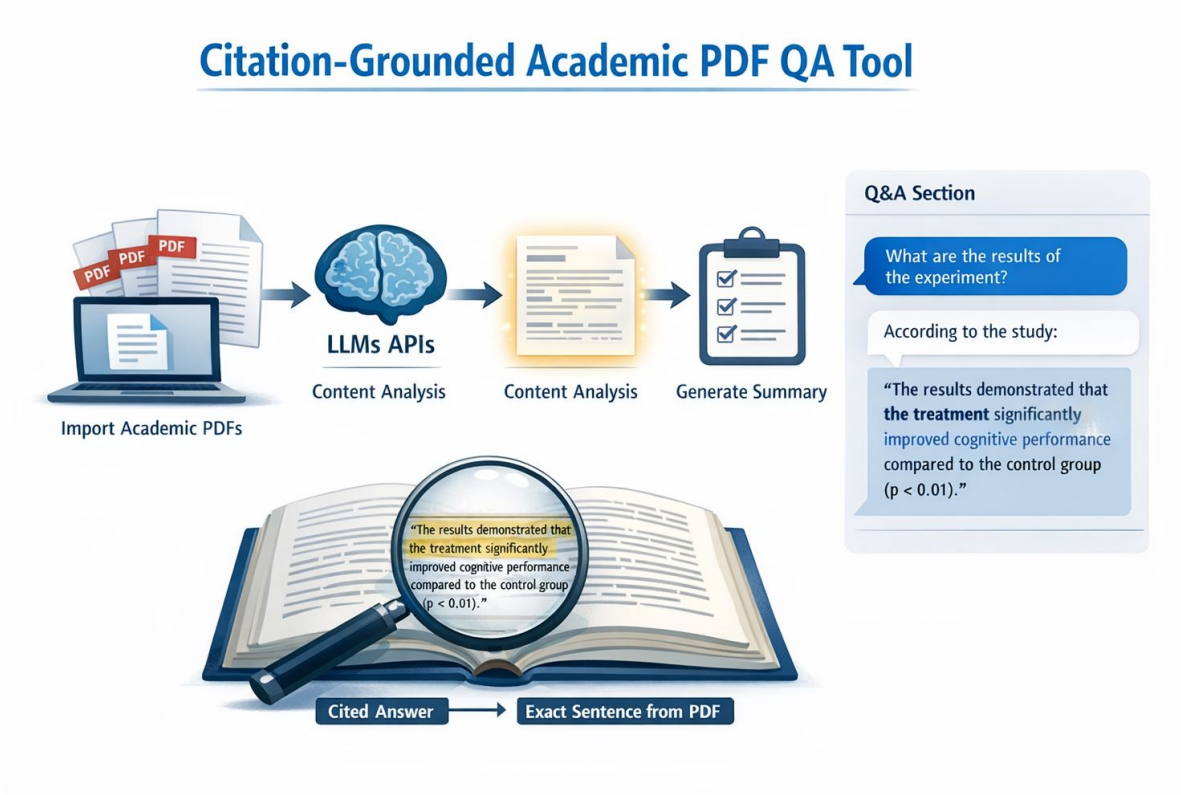
1. Researchers spend significant time reading long academic papers
2. Existing LLM tools often:
   a. Hallucinate answers
   b. Do not provide exact citations
3. Need a reliable, citation-grounded AI assistant
4. Compare multiple LLM models to evaluate answer consistency



Project Scope

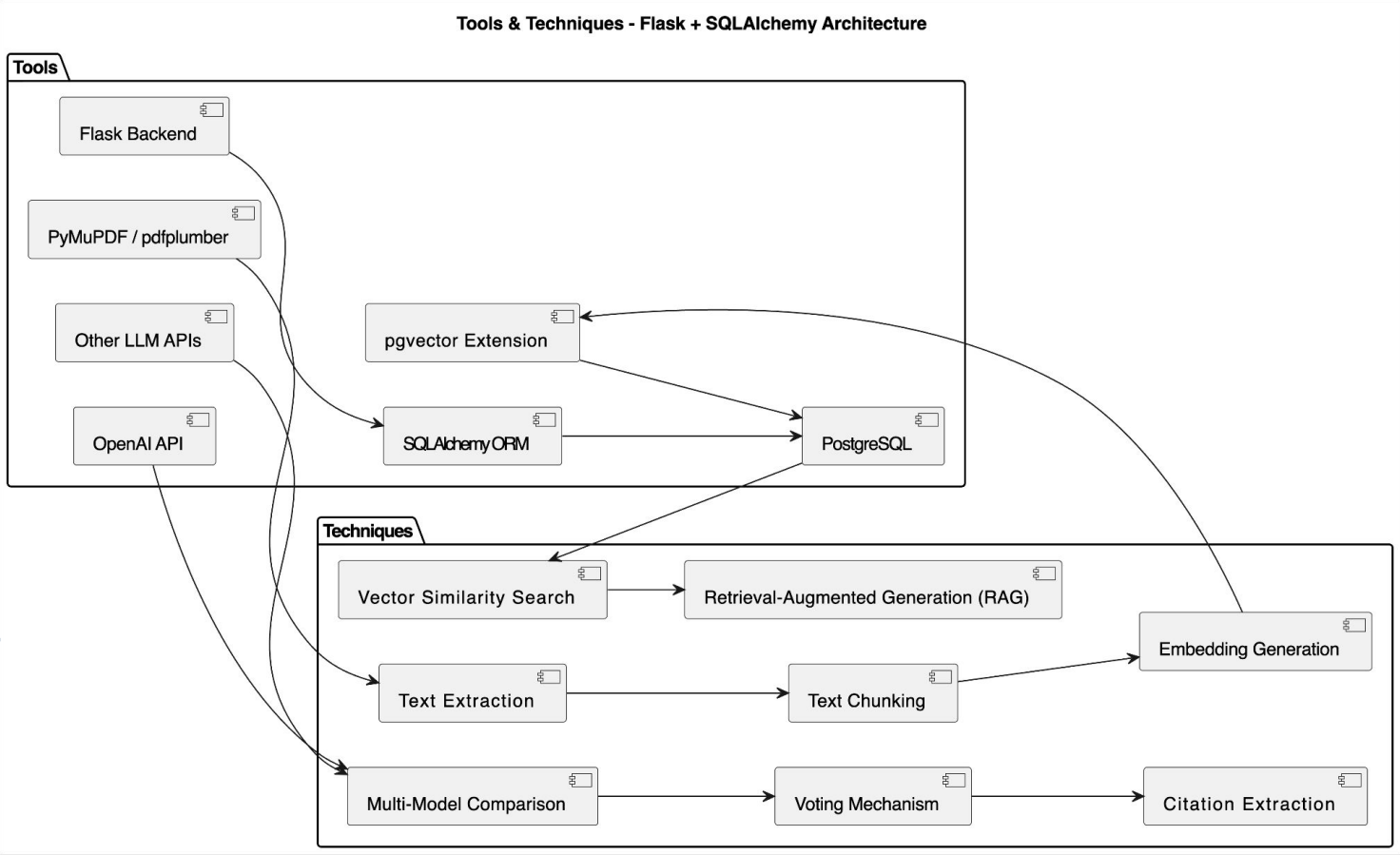
Citation-Grounded Academic PDF QA Tool will:

1. Imports academic PDF papers
2. Uses multiple LLM APIs to analyze content
3. Generates summaries
4. Supports Q&A grounded strictly in the paper
5. Returns exact citation sentences from the PDF

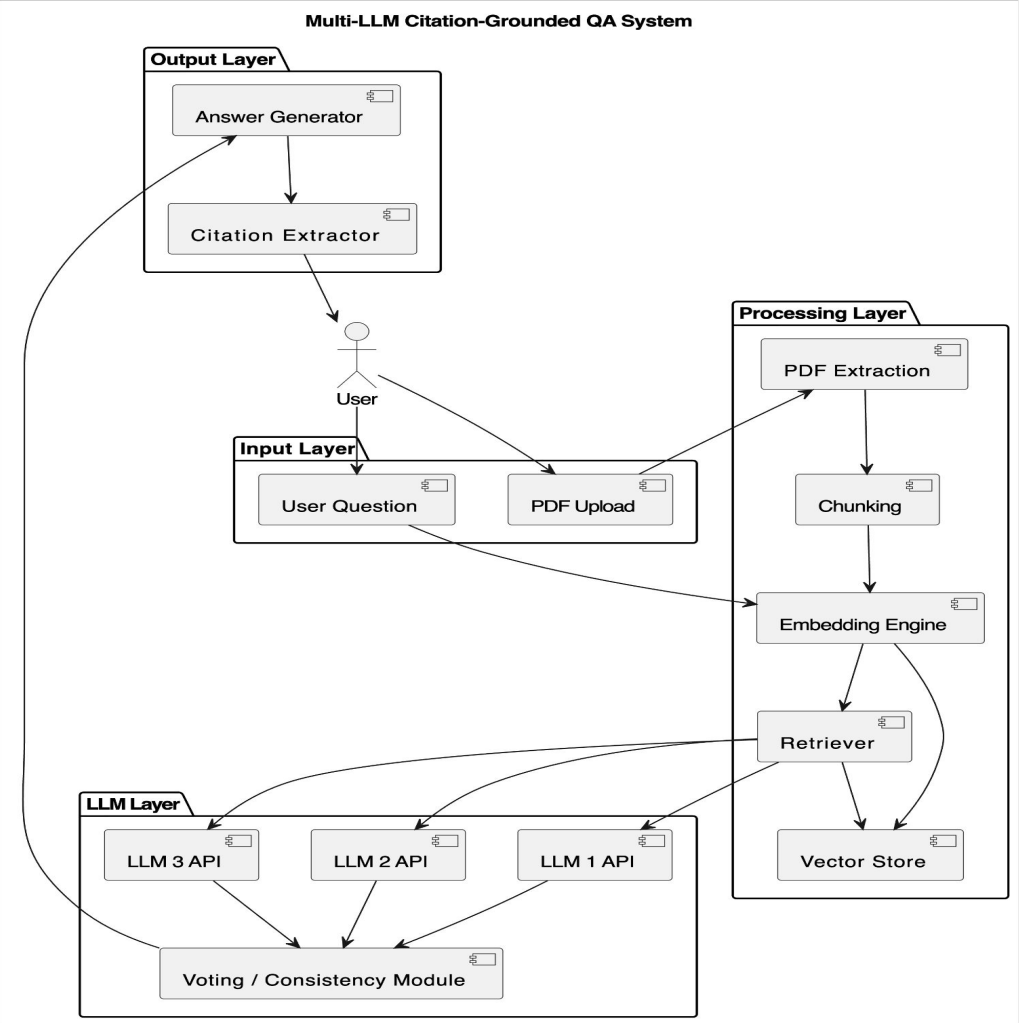


Tools & Techniques

1. Flask-based Python backend
2. SQLAlchemy ORM
3. Large Language Model APIs
4. PostgreSQL with pgvector extension
5. Retrieval-Augmented Generation (RAG)
6. Citation extraction and grounding techniqu



System Architecture



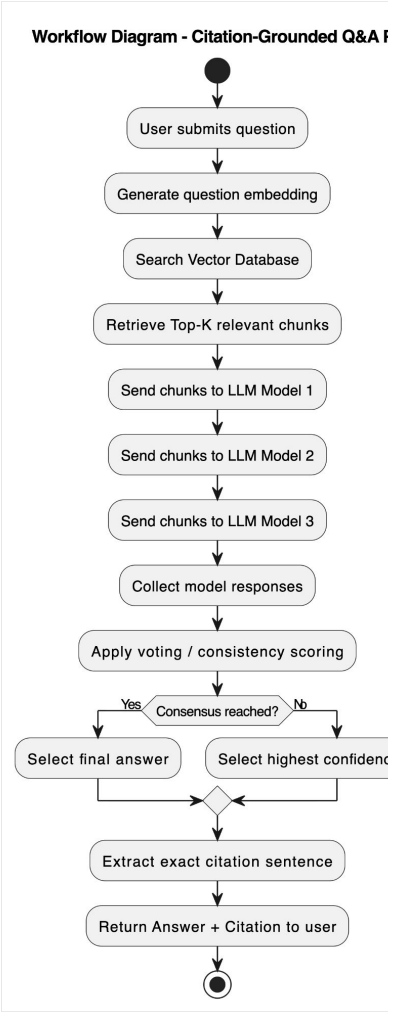
Workflow Diagram

Citation-Grounded Q&A Process (3 steps)

1. The user submits a question, and the system converts it into an embedding.

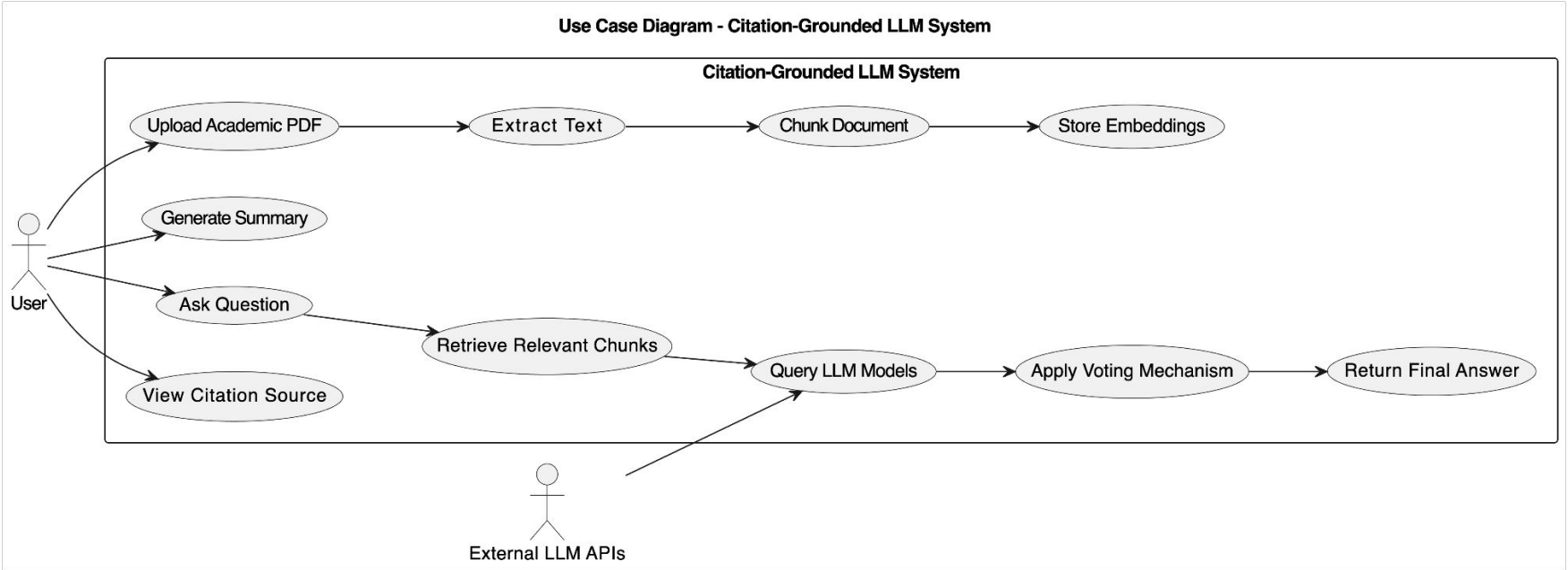
2. The system searches the vector database, retrieves the most relevant chunks, and sends them to multiple LLM models.

3. The system compares the responses, selects the best answer, extracts the exact citation, and returns the final answer with citation.



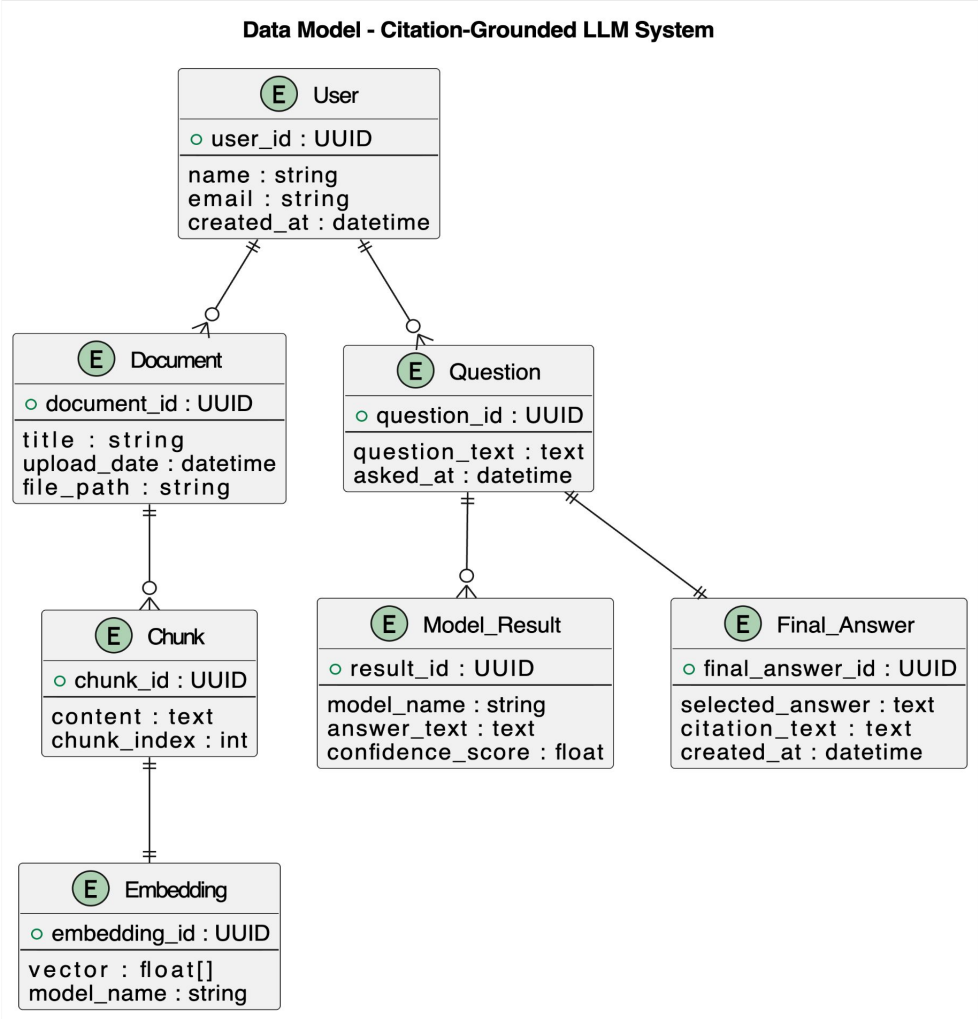
Use Case Diagram

1. The user uploads a PDF. The system extracts, chunks, and embeds the text in PostgreSQL.
2. For summaries or questions, it retrieves relevant chunks, queries multiple LLMs, applies voting, and returns the most reliable answer with exact citation.



Data Model

1. Users can upload multiple documents and ask multiple questions.
2. Each document is split into chunks with embeddings.
3. Each question generates multiple model results.
4. The system selects one final answer with exact citation.



Implementation Plan

Implementation Roadmap

Phase 1 → Phase 2 → Phase 3 → Phase 4 → Phase 5

Phase 1 – PDF Processing

- Extract
- Chunk
- Embed

Deliverable: Searchable Index

Phase 2 – Retrieval + Q&A

- Top-K Retrieval
- Multi-LLM
- Voting

Deliverable: Grounded Q&A

Phase 3 – Citation Extraction

- Exact sentence match
- No hallucination

Deliverable: Citation Output

Phase 4 – Evaluation

- Consistency
- Accuracy
- Latency

Deliverable: Comparison Report

Phase 5 – UI / Demo

- Upload
- Ask question
- Show citation

Deliverable: Working System

Evaluation

1. Develop a structured framework to evaluate multi-LLM citation-grounded QA.

2. Assess answer correctness and evidence support.

3. Compare models under the same retrieval context.

4. Study voting mechanisms and quantify hallucination reduction.

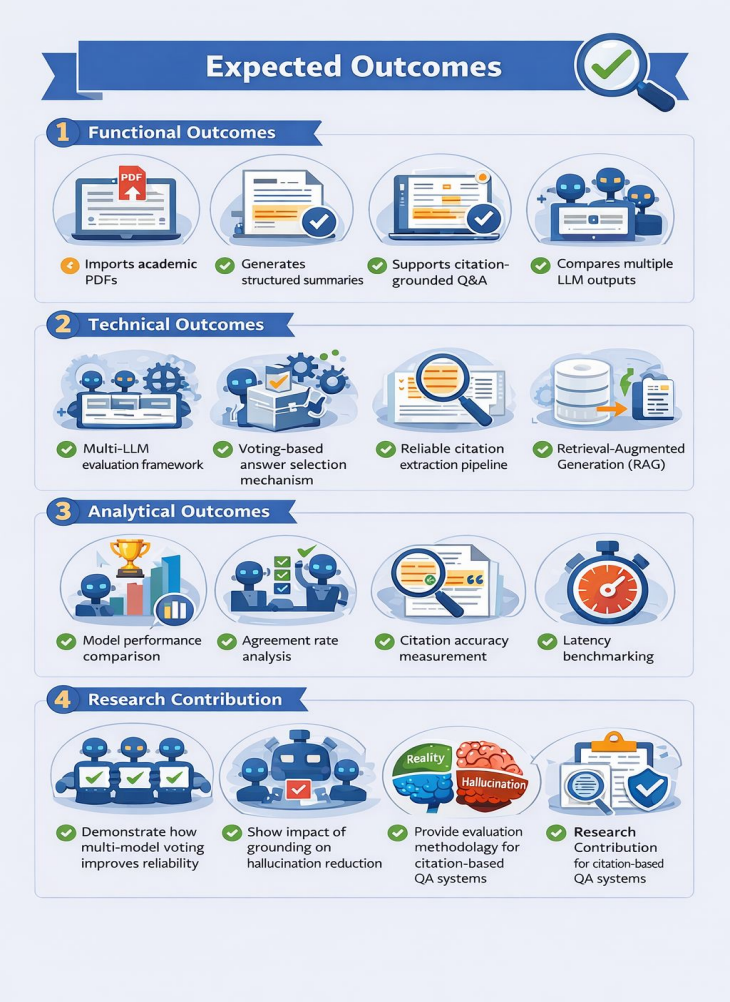
Expected Outcomes

1. Build a multi-LLM system for summarizing PDFs and answering citation-grounded questions.

2. Integrate RAG, multi-model querying, and voting-based selection.

3. Provide measurable evaluation (accuracy, agreement, citation precision, latency).

4. Demonstrate improved reliability over single-model approaches.



Thank you!